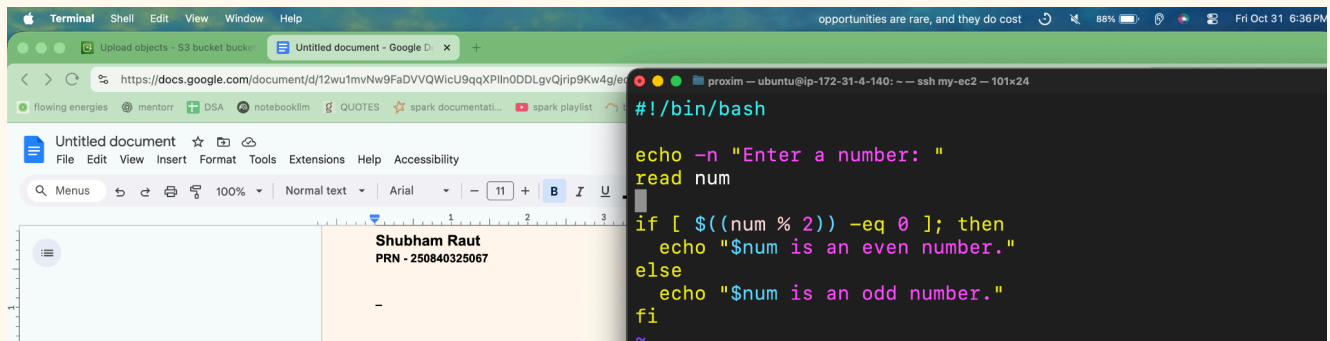


Q1.

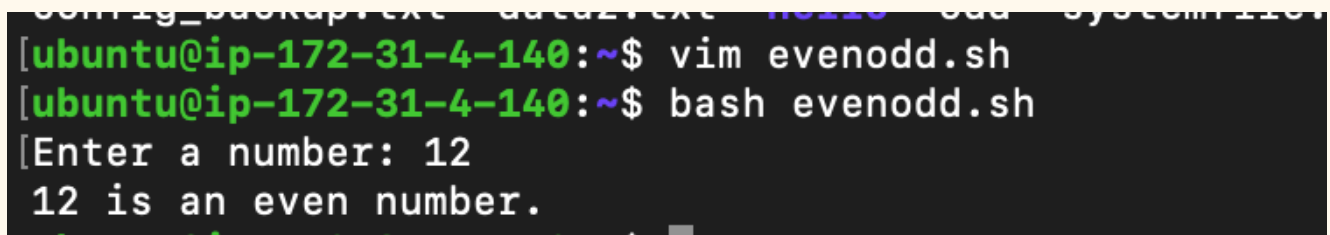
1. check if the number is odd or even



```
#!/bin/bash

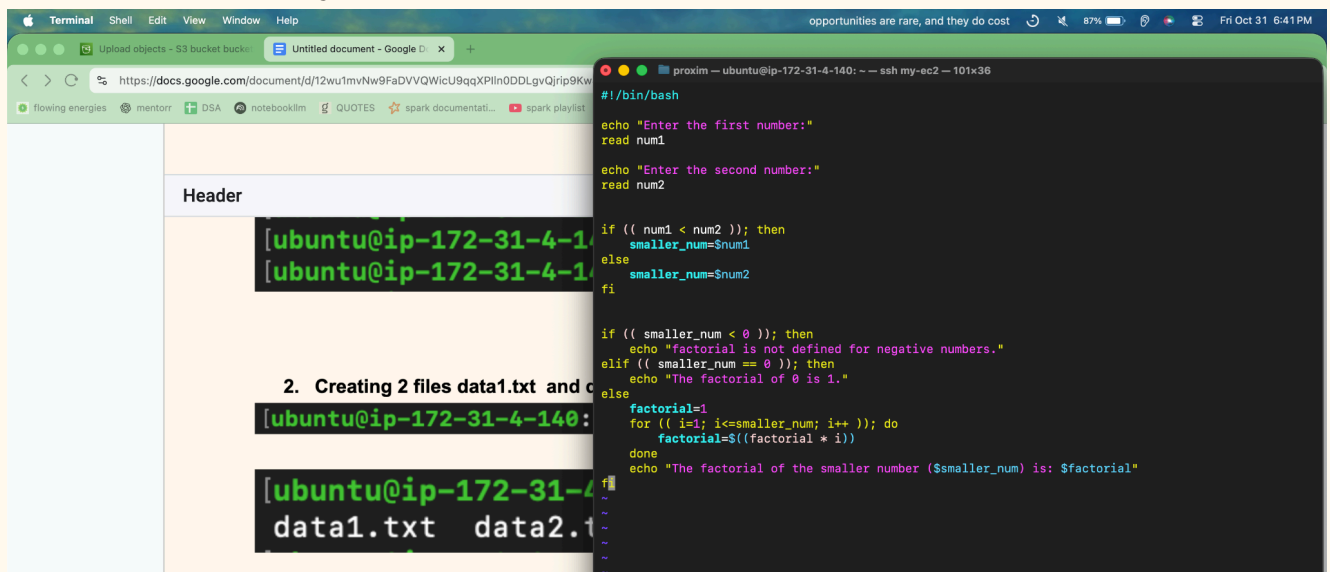
echo -n "Enter a number: "
read num

if [  $((num \% 2)) -eq 0$  ]; then
    echo "$num is an even number."
else
    echo "$num is an odd number."
fi
```



```
[ubuntu@ip-172-31-4-140:~$ vim evenodd.sh
[ubuntu@ip-172-31-4-140:~$ bash evenodd.sh
Enter a number: 12
12 is an even number.
```

2. Calculate and display the factorial of the smaller number



```
#!/bin/bash

echo "Enter the first number:"
read num1

echo "Enter the second number:"
read num2

if (( num1 < num2 )); then
    smaller_num=num1
else
    smaller_num=num2
fi

if (( smaller_num < 0 )); then
    echo "factorial is not defined for negative numbers."
elif (( smaller_num == 0 )); then
    echo "The factorial of 0 is 1."
else
    factorial=1
    for (( i=1; i<=smaller_num; i++ )); do
        factorial=$((factorial * i))
    done
    echo "The factorial of the smaller number ($smaller_num) is: $factorial"
```

```
[ubuntu@ip-172-31-4-140:~$ bash factorial.sh
Enter the first number:
12
Enter the second number:
12
The factorial of the smaller number (12) is: 479001600
```

Q2.

1. Create directory named "student_data"

```
[ubuntu@ip-172-31-4-140:~$ mkdir tmp
[ubuntu@ip-172-31-4-140:~$ cd tmp
[ubuntu@ip-172-31-4-140:~/tmp$ mkdir student_data
```

2. Creating 2 files data1.txt and data2.txt

```
[ubuntu@ip-172-31-4-140:~/tmp$ touch data1.txt data2.txt
```

```
[ubuntu@ip-172-31-4-140:~/tmp$ ls
data1.txt  data2.txt  student_data
```

3. Rename the file to the config_backup.txt

```
[ubuntu@ip-172-31-4-140:~/tmp$ mv data1.txt config_backup.txt
```

4. Change permissions of config_backup.txt to read only to owner

```
[ubuntu@ip-172-31-4-140:~$ sudo chmod 400 config_backup.txt
[ubuntu@ip-172-31-4-140:~$ ls
config_backup.txt  hello  odd  tmp
ubuntu@ip-172-31-4-140:~$ ls -la
total 68
drwxr-x--- 9 ubuntu ubuntu 4096 Oct 31 12:48 .
drwxr-xr-x 3 root   root   4096 Oct 29 05:13 ..
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 31 05:56 .bash
-rw----- 1 ubuntu ubuntu 1248 Oct 31 11:35 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Mar 31  2024 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Oct 29 05:22 .cache
drwx----- 3 ubuntu ubuntu 4096 Oct 29 05:31 .config
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 29 05:23 .local
-rw-r--r-- 1 ubuntu ubuntu  807 Mar 31  2024 .profile
drwx----- 2 ubuntu ubuntu 4096 Oct 29 05:23 .ssh
-rw-r--r-- 1 ubuntu ubuntu    0 Oct 31 12:48 .sudo_as_admin_successful
-rw----- 1 ubuntu ubuntu 9097 Oct 31 12:42 .viminfo
-r----- 1 ubuntu ubuntu    0 Oct 31 12:47 config_backup.txt
```

```
[ubuntu@ip-172-31-4-140:~/tmp$ ls -la
total 16
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 31 12:50 .
drwxr-x--- 9 ubuntu ubuntu 4096 Oct 31 12:48 ..
-rw-rw-r-- 1 ubuntu ubuntu    0 Oct 31 12:42 data1.txt
-rw-rw-r-- 1 ubuntu ubuntu    0 Oct 31 12:42 data2.txt
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 31 12:42 student_data
```

5. Compressed tar of the entire student_data

```
ubuntu@ip-172-31-4-140:~/tmp$ tar -cvzf student_data student_data.tar.gz home/ubuntu/
```

```
[ubuntu@ip-172-31-4-140:~/tmp$ ls -la
total 16
drwxrwxr-x 3 ubuntu ubuntu 4096 Oct 31 12:50 .
drwxr-x--- 9 ubuntu ubuntu 4096 Oct 31 12:48 ..
-rw-rw-r-- 1 ubuntu ubuntu   0 Oct 31 12:42 data1.txt
-rw-rw-r-- 1 ubuntu ubuntu   0 Oct 31 12:42 data2.txt
drwxrwxr-x 2 ubuntu ubuntu 4096 Oct 31 12:42 student_data
-rw-rw-r-- 1 ubuntu ubuntu  45 Oct 31 12:52 student_data.tar.gz
```

1. Creating a user with a name exam-user

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings
- Root access management

Access reports

- Access Analyzer
 - Resource analysis [New](#)
 - Unused access
 - Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies

User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

View user

Users (3) [info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID
☐	exam-user	/	0	-	-	-	-	-
☐	payal	/	0	-	-	-	-	-
☐	shubham	/	1	26 days ago	-	-	-	Active - AKIAZWC222U...

2. Creating a group with exam-group

Brave File Edit View History Bookmarks Profiles Tab Window Help opportunities are rare, and they do cost 95% Fri Oct 31 5:38 PM

exam-group | IAM | Global x +

https://us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/groups/details/exam-group?section=users

aws Search [Option+S]

Account ID: 6659-1043-3087 SHUBHAM-CDAC

IAM > User groups > exam-group

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Access reports

Access Analyzer

Resource analysis New

Unused access

Analyzer settings

Credential report

Organization activity

Service control policies

Resource control policies

exam-group user group created. View group X

exam-group Info Delete

Summary Edit

User group name exam-group

Creation time October 31, 2025, 17:37 (UTC+05:30)

ARN arn:aws:iam::665910433087:group/exam-group

Users (1) Permissions Access Advisor

Users in this group (1)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

Remove Add users

1 5 minutes ago

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exam-group | IAM | Global x +

https://us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/groups/details/exam-group?section=users

aws Search [Option+S]

Account ID: 6659-1043-3087 SHUBHAM-CDAC

IAM > User groups > exam-group

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Root access management

Access reports

Access Analyzer

Resource analysis New

Unused access

Analyzer settings

Credential report

Organization activity

Service control policies

Resource control policies

exam-group user group created. View group X

exam-group Info Delete

Summary Edit

User group name exam-group

Creation time October 31, 2025, 17:37 (UTC+05:30)

ARN arn:aws:iam::665910433087:group/exam-group

Users (1) Permissions Access Advisor

Users in this group (1)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

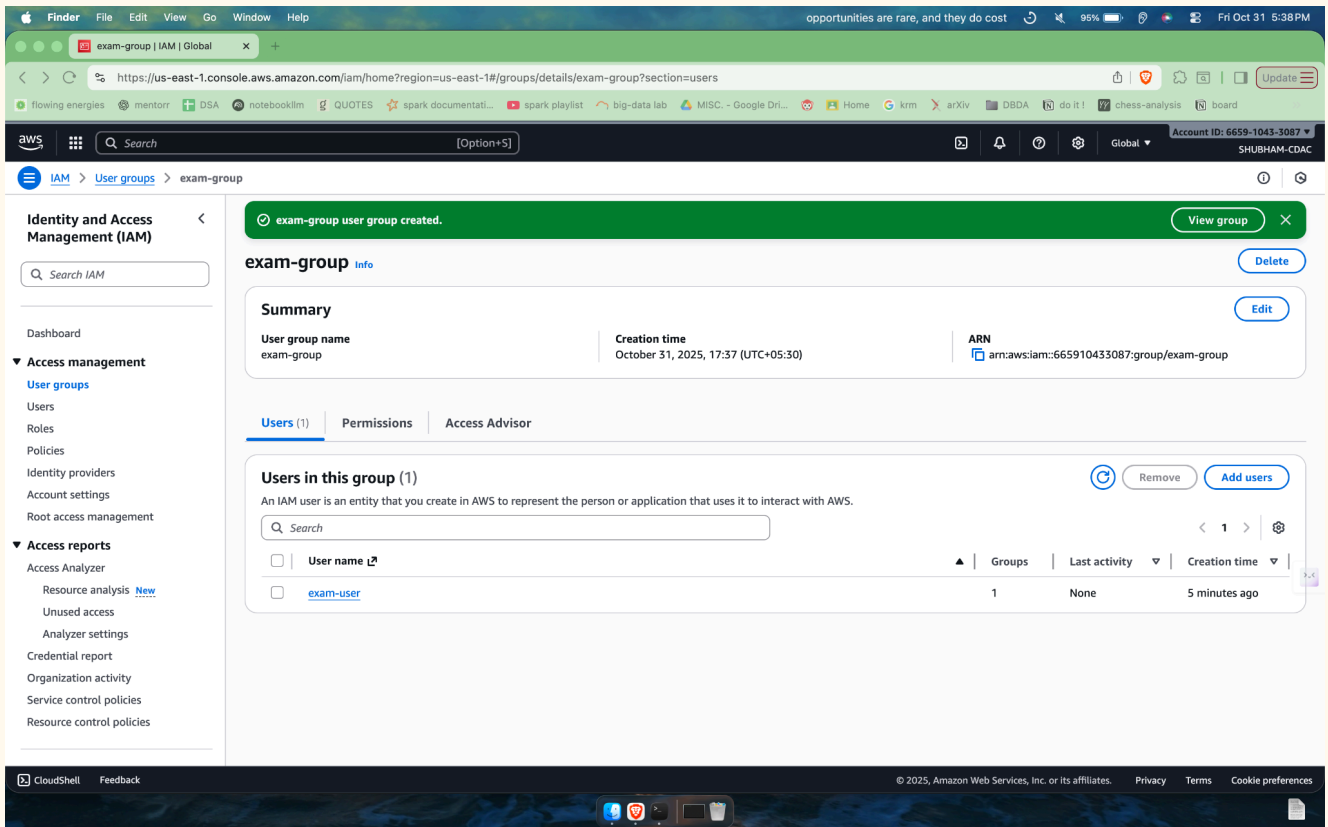
Search

Remove Add users

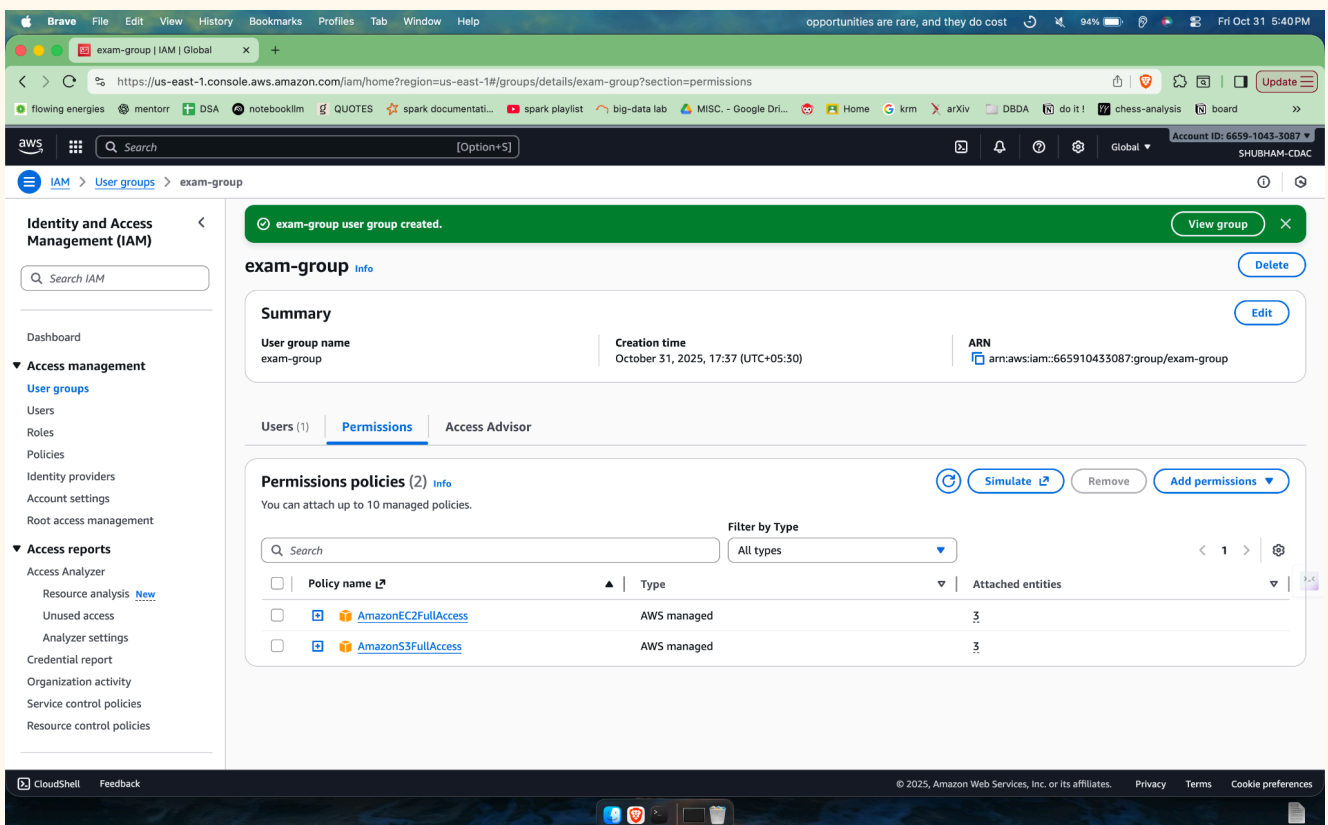
1 5 minutes ago

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3. Attaching policies EC2FullAccess and S3FullAccess to the user and the group



Stopping the instance named exam-instance

The screenshot shows the AWS Management Console for the 'us-east-1' region. The 'Instances' page is active, displaying a table of EC2 instances. The instance 'exam-instance' (ID: i-00cbdc8f06c538075) is selected, and its state is 'Stopping'. A dropdown menu is open under the 'Actions' column, showing options like 'Force stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'. Below the table, the details for 'exam-instance' are shown, including its instance ID, IP addresses, and state.

Successfully initiated stopping of i-00cbdc8f06c538075

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Public IPv4 DNS
my-instance	i-0586f2cd887f9e2b	Running	t2.micro	2/2 checks passed	ec2-54-87-95-39.com...
exam-instance	i-00cbdc8f06c538075	Stopping	t2.micro	Initializing	ec2-54-242-172-77.co...

i-00cbdc8f06c538075 (exam-instance)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-00cbdc8f06c538075

IPV6 address -

Hostname type IP name: ip-172-31-17-120.ec2.internal

Answer private resource DNS name IPv4 (A)

Public IPv4 address 54.242.172.77 | open address

Instance state Stopping

Private IP DNS name (IPv4 only) ip-172-31-17-120.ec2.internal

Instance type t2.micro

Private IPv4 addresses 172.31.17.120

Public DNS ec2-54-242-172-77.compute-1.amazonaws.com | open address

Elastic IP addresses -

Starting the stopped instance once again

The screenshot shows the AWS Management Console for the 'us-east-1' region. The 'Launch an instance' page is active, displaying a success message: 'Successfully initiated launch of instance (i-00cbdc8f06c538075)'. Below the message, the 'Launch log' shows the progress of the launch process, including 'Initializing requests', 'Creating security groups', 'Creating security group rules', and 'Launch initiation', all of which are marked as 'Succeeded'. The 'Next Steps' section provides suggestions for what to do next with the instance, such as 'Create billing usage alerts', 'Connect to your instance', 'Connect an RDS database', 'Create EBS snapshot policy', 'Manage detailed monitoring', 'Create Load Balancer', 'Create AWS budget', and 'Manage CloudWatch alarms'.

Success

Successfully initiated launch of instance (i-00cbdc8f06c538075)

Launch log

Initializing requests Succeeded

Creating security groups Succeeded

Creating security group rules Succeeded

Launch initiation Succeeded

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Create Load Balancer

Create AWS budget

Manage CloudWatch alarms

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Instances | EC2 | us-east-1 x +

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instances:

flowing energies mentorr DSA notebooklm QUOTES spark documentati... spark playlist big-data lab MISC. - Google Dri... Home G krm arXiv DBDA do it! chess-analysis board

aws Search [Option+S] United States (N. Virginia) Account ID: 6659-1043-3087 SHUBHAM-CDAC

EC2 > Instances

EC2 Dashboard EC2 Global View Events

▼ Instances Instances Instance Types Launch Templates Spot Requests Savings Plans Reserved Instances Dedicated Hosts Capacity Reservations Capacity Manager New

▼ Images AMIs AMI Catalog

▼ Elastic Block Store Volumes Snapshots Lifecycle Manager

▼ Network & Security Security Groups Elastic IP

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Successfully initiated starting of i-00cbdc8f06c538075

Instances (1/2) Info

Find instance by attribute or tag (case-sensitive) All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☐	my-instance	i-0586ff2cd887f9e2b	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-54-87-95-39.comp...
☒	exam-instance	i-00cbdc8f06c538075	Pending	t2.micro	-	View alarms +	us-east-1c	-

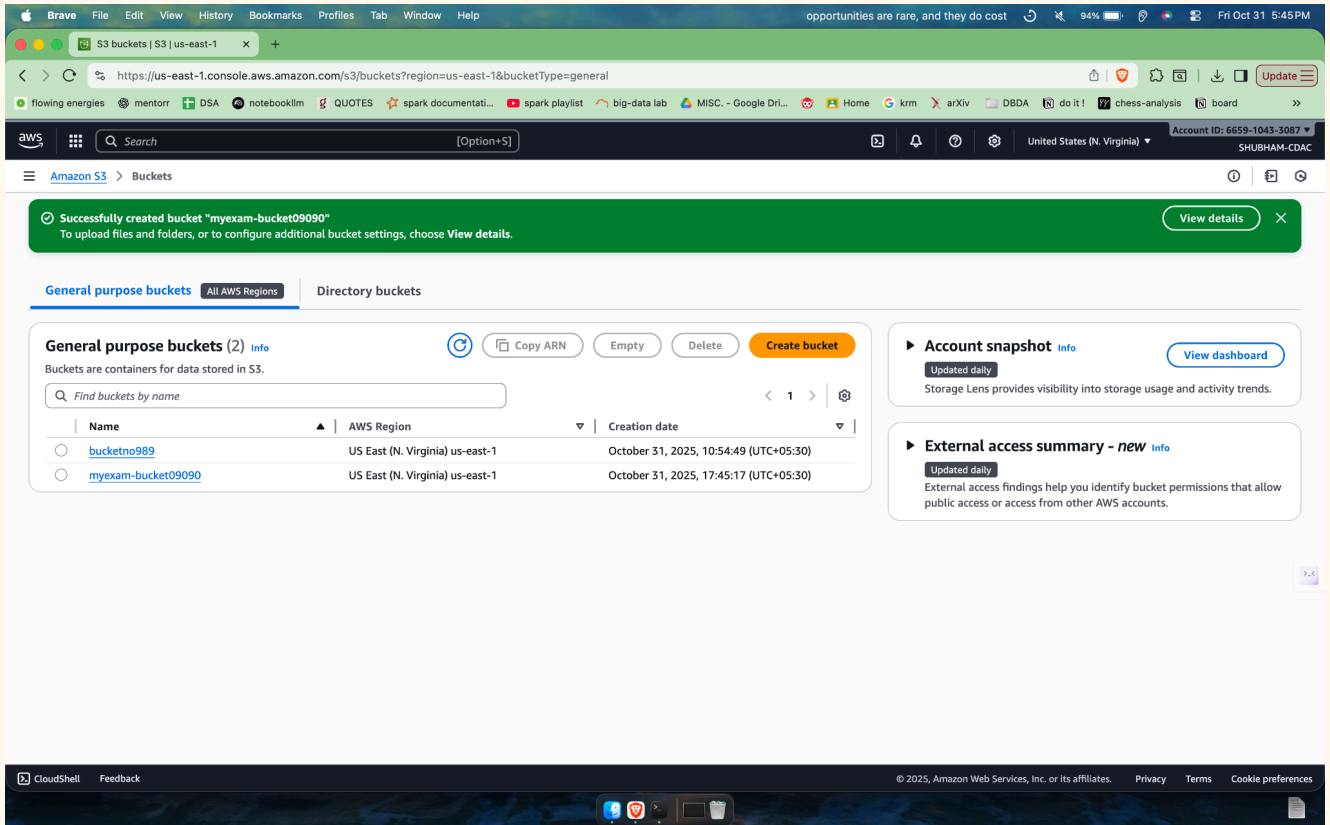
i-00cbdc8f06c538075 (exam-instance)

Details Status and alarms Monitoring Security Networking Storage Tags

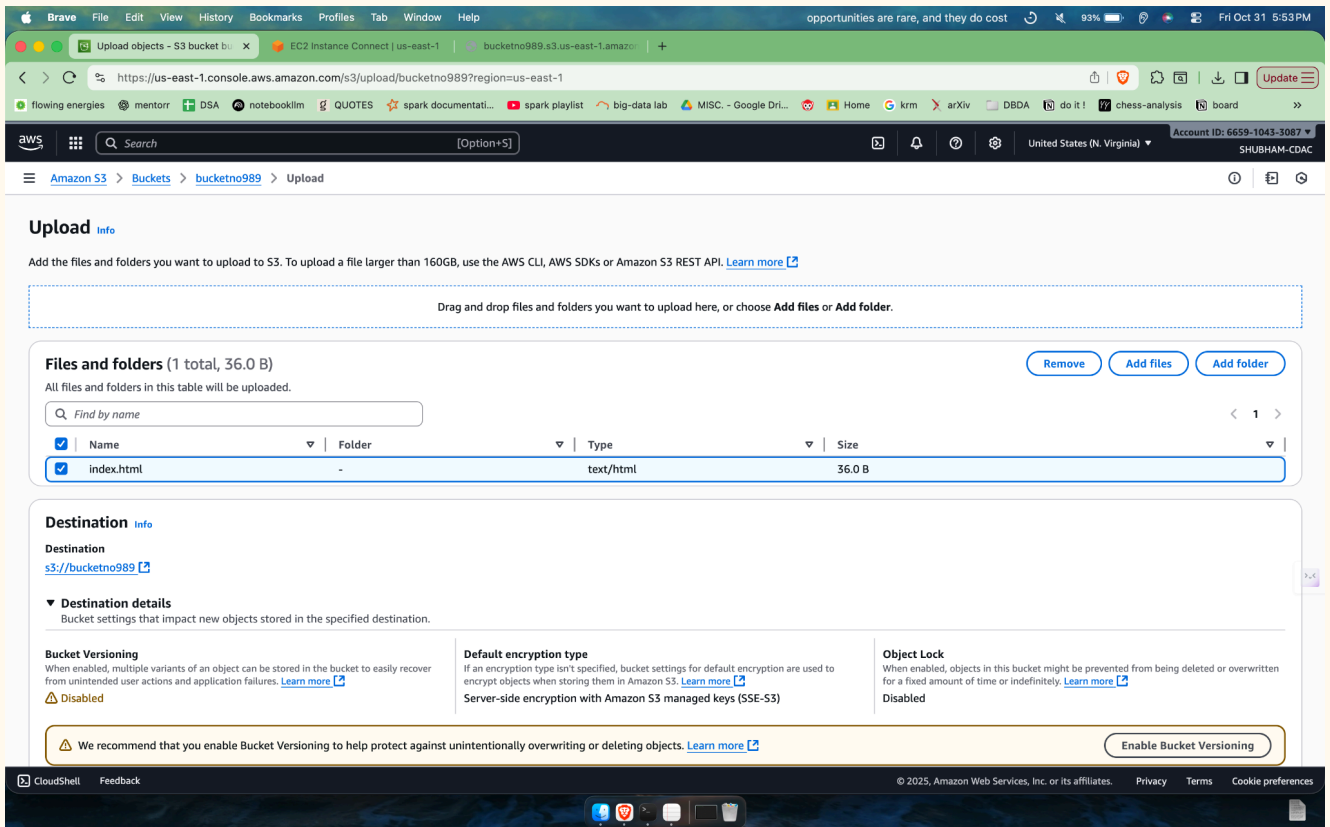
▼ Instance summary Info

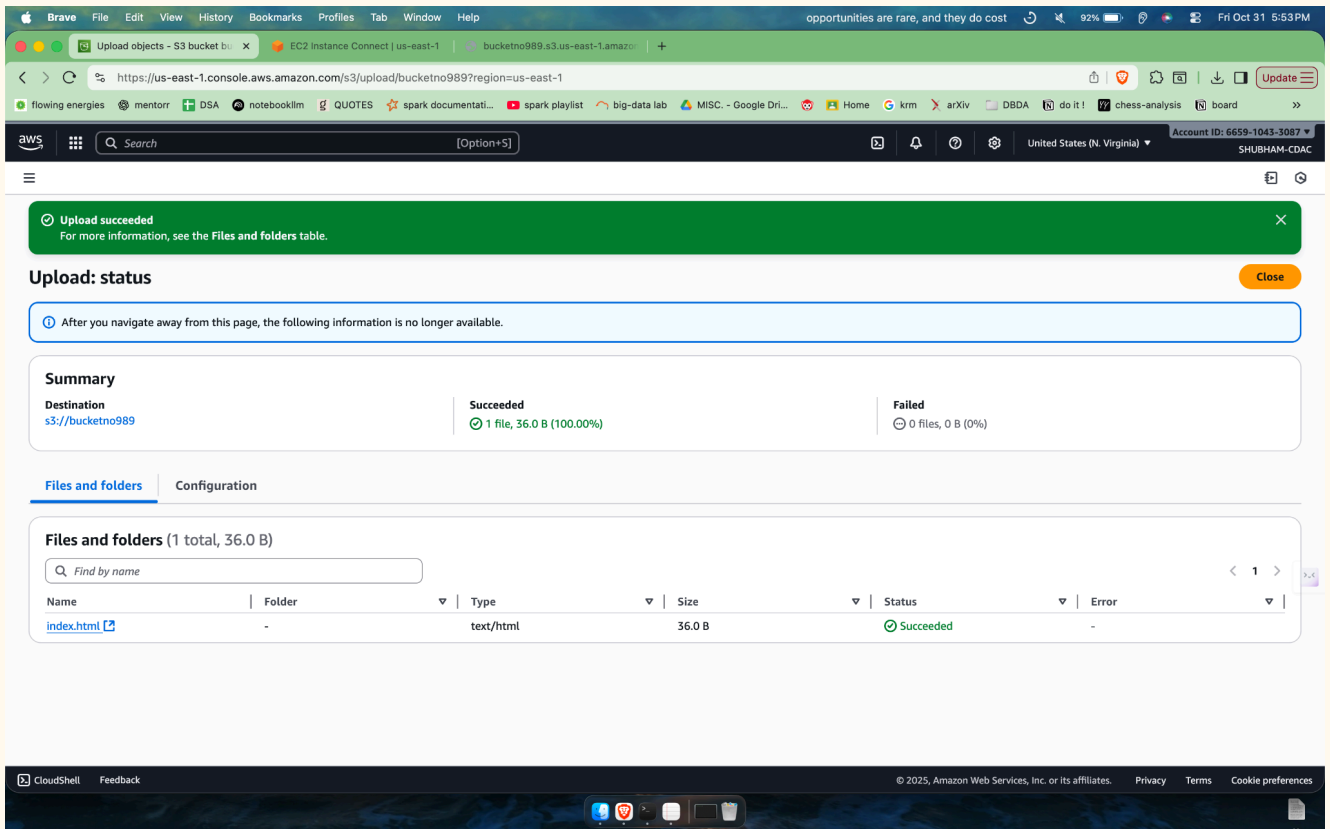
Instance ID i-00cbdc8f06c538075	Public IPv4 address -	Private IPv4 addresses 172.31.17.120
IPv6 address -	Instance state Pending	Public DNS -
Hostname type IP name: ip-172-31-17-120.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-17-120.ec2.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t2.micro	

Creating bucket named myexam-bucket09090

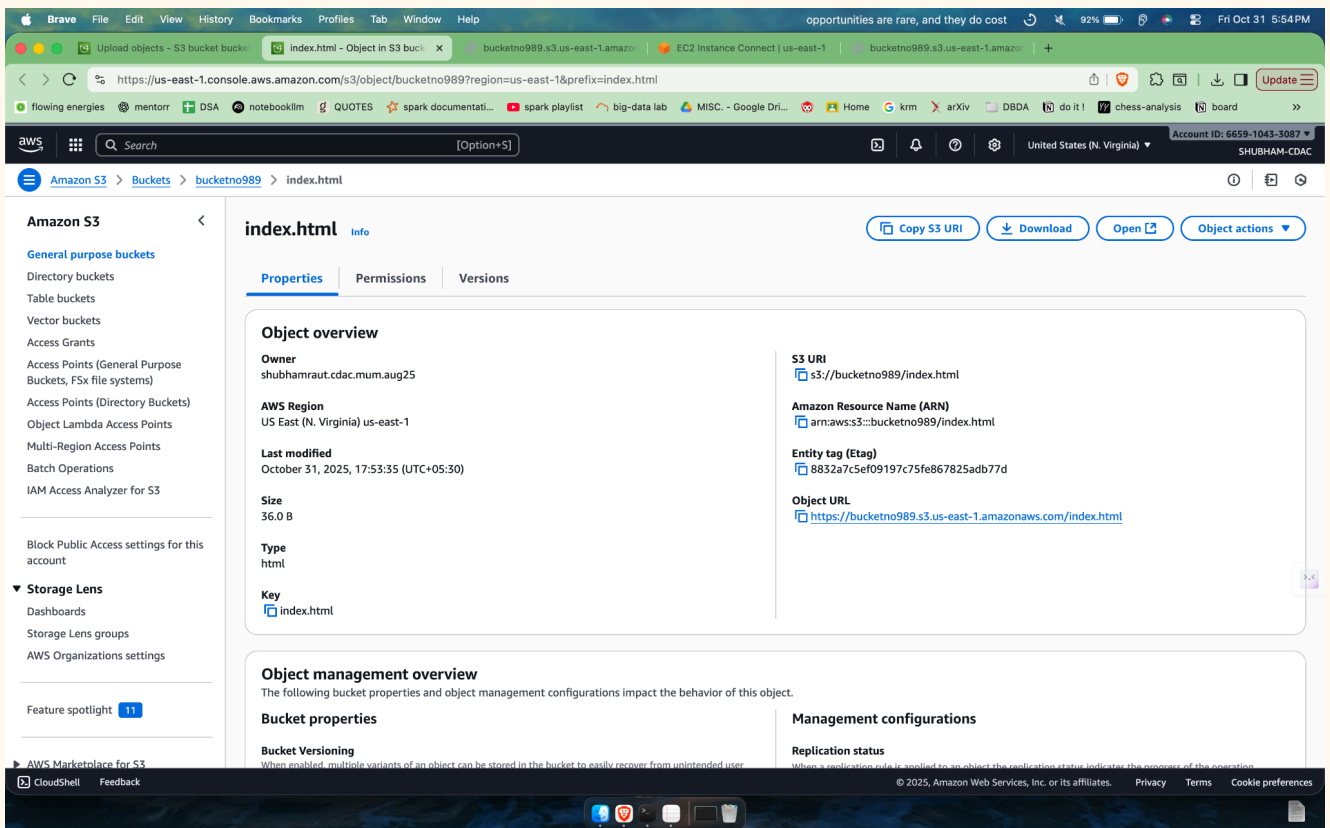


Uploading the file index.html to the bucket





Accessing the object bucket url for the image



Viewing the file / page accessible as public on object url

